

How the Compliance Score Is Generated

A step-by-step walkthrough of how `compliance_matrix.yaml` values are used by `scoring_engine.py` to compute a 0–100 priority score for each email, with three worked examples.

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scoring formula

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YAML categories
with weights

2

multipliers applied

5

priority bands
0→100 scale

1. The compliance_matrix.yaml File

All scoring parameters live in **config/compliance_matrix.yaml**. The scoring engine reads this file on startup and uses it for every calculation. Changing a weight or threshold in the YAML takes effect on the next run — no code changes needed.

Category weights (the key input)

Each of the 6 compliance categories has a **base_weight** from 0–10 reflecting how serious that violation is considered to be.

```
categories:
market_manipulation:
base_weight: 10 # Most severe – direct market crime
bribery:
base_weight: 9 # Very severe – improper payments
secrecy:
base_weight: 8 # Severe – leaking non-public information
employee_ethics:
base_weight: 7 # Serious – conflicts of interest
change_in_communication:
base_weight: 6 # Concerning – evading surveillance
complaints:
base_weight: 4 # Lower – client/regulatory grievances
```

Multipliers

Multipliers are applied on top of the base weight total to push the score higher when additional risk factors are present.

```
scoring:
multipliers:
multi_category_bonus: 1.20 # +20% when 2 or more categories flagged
high_confidence_bonus: 1.15 # +15% when AI confidence >= 0.80
max_raw_score: 10 # Ceiling used to normalise raw score to 0-100
```

Priority bands and alert levels

After the 0–100 score is computed it is mapped to a named band. The alert level is assigned separately based on the AI's confidence value.

```
priority_bands:
critical: 80 # score >= 80 -> CRITICAL
high: 60 # score >= 60 -> HIGH
medium: 40 # score >= 40 -> MEDIUM
low: 0 # score >= 0 -> LOW

confidence_thresholds:
high: 0.80 # AI confidence >= 0.80 -> Alert: HIGH
medium: 0.50 # AI confidence >= 0.50 -> Alert: MEDIUM
low: 0.00 # AI confidence < 0.50 -> Alert: LOW
```

2. The Scoring Formula (4 Steps)

The scoring engine in `src/scoring_engine.py` executes these four steps in sequence for every non-compliant email.

1

Sum the base weights

Read `base_weight` from the YAML for each detected category. Add them together to get the raw score.

2

Apply multi-category bonus (if 2+ categories)

If the AI detected 2 or more categories, multiply raw by `multi_category_bonus` (1.20 from YAML). This reflects that an email breaking multiple rules is more serious.

3

Apply high-confidence bonus (if confidence ≥ 0.80)

If the AI's confidence is ≥ 0.80 (high threshold from YAML), multiply by `high_confidence_bonus` (1.15). Rewarding high-certainty findings.

4

Normalise to 0-100

`priority_score = min(100, int(raw / max_raw_score * 100))`. `max_raw_score` is 10 in the YAML. So a raw score of 10 = 100/100. The `min(100,...)` caps it so it never exceeds 100.

The formula in code (`scoring_engine.py`)

```
# Step 1: sum weights from YAML
raw_score = 0.0
for cat_id in categories:
    weight = config['categories'][cat_id]['base_weight']
    raw_score += weight

# Step 2: multi-category bonus
if len(categories) >= 2:
    raw_score *= config['scoring']['multipliers']['multi_category_bonus'] # 1.20

# Step 3: high-confidence bonus
if confidence >= config['scoring']['confidence_thresholds']['high']: # 0.80
    raw_score *= config['scoring']['multipliers']['high_confidence_bonus'] # 1.15

# Step 4: normalise to 0-100
priority_score = min(100, int(raw_score / config['scoring']['max_raw_score'] * 100))
```

3. Worked Examples

Three real emails from the test dataset, showing exactly how the YAML values feed into the formula at each step.

Example 1 — Market Manipulation (single category, high confidence)

Email: "coordinate our buy orders around 09:31 we can push the price up 4-5%"

Step	YAML value used	Calculation	Running total
1 — Base weight	market_manipulation: base_weight: 10	raw = 10	raw = 10.0
2 — Multi-category bonus	Only 1 category detected	Not applied (need >= 2 categories)	raw = 10.0
3 — High-confidence bonus	high_confidence_bonus : 1.15 high threshold: 0.80 AI confidence = 0.96	0.96 >= 0.80 YES raw = 10.0 × 1.15	raw = 11.5
4 — Normalise	max_raw_score: 10	min(100, int(11.5/10×100)) = min(100, 115) = 100	score = 100
Band assignment	critical: 80	100 >= 80 -> CRITICAL	CRITICAL
Alert level	high threshold: 0.80 confidence = 0.96	0.96 >= 0.80 -> HIGH alert	Alert: HIGH

100/100

Final result: priority_score = 100/100 | Band: CRITICAL | Alert: HIGH

Example 2 — Secrecy + Change in Communication (two categories)

Email: "Don't put anything about the Vega deal in email. Use Signal from now on. Delete this thread."

Step	YAML value used	Calculation	Running total
1 — Base weights	secrecy: base_weight: 8 change_in_communication: base_weight: 6	raw = 8 + 6	raw = 14.0
2 — Multi-category bonus	multi_category_bonus: 1.20 (2 categories found)	2 >= 2 YES raw = 14.0 × 1.20	raw = 16.8
3 — High-confidence bonus	high threshold: 0.80 AI confidence = 0.91	0.91 >= 0.80 YES raw = 16.8 × 1.15	raw = 19.32

4 — Normalise	max_raw_score: 10	$\min(100, \text{int}(19.32/10 \times 100)) = \min(100, 193) = 100$	score = 100
Band assignment	critical: 80	$100 \geq 80 \rightarrow \text{CRITICAL}$	CRITICAL
Alert level	high threshold: 0.80 confidence = 0.91	$0.91 \geq 0.80 \rightarrow \text{HIGH alert}$	Alert: HIGH

100/100

Final result: priority_score = 100/100 | Band: CRITICAL | Alert: HIGH (multi-category bonus boosted raw from 14 to 19.32)

Example 3 — Complaints only (medium confidence)

Email: "I am formally complaining about three unauthorised trades. If unresolved I will escalate to the FCA."

Step	YAML value used	Calculation	Running total
1 — Base weight	complaints: base_weight: 4	raw = 4	raw = 4.0
2 — Multi-category bonus	Only 1 category detected	Not applied	raw = 4.0
3 — High-confidence bonus	high threshold: 0.80 AI confidence = 0.72	0.72 < 0.80 NO Not applied	raw = 4.0
4 — Normalise	max_raw_score: 10	min(100, int(4.0/10×100)) = min(100, 40) = 40	score = 40
Band assignment	medium: 40 high: 60	40 >= 40 but < 60 -> MEDIUM	MEDIUM
Alert level	medium threshold: 0.50 confidence = 0.72	0.72 >= 0.50 -> MEDIUM alert	Alert: MEDIUM

40/100

Final result: priority_score = 40/100 | Band: MEDIUM | Alert: MEDIUM (low base weight + confidence below 0.80 = no bonuses)

Example 4 — Compliant email (zero score)

Email: "Please find attached the Q1 settlement reconciliation. Deadline for corrections is 17:00 Friday."

The AI returns is_compliant=true and categories=[]. The scoring engine short-circuits immediately: priority_score = 0 | band = COMPLIANT | alert = NONE No YAML weights are read. No formula is executed.

4. Full Category Weight Reference

All six categories from `compliance_matrix.yaml` with their weights, description, and the score they produce when detected alone at 100% confidence.

Category ID	Label	Weight	Alone score (conf 1.0)	Band
market_manipulation	Market Manipulation	10	$\min(100, \text{int}(10 \times 1.15 / 10 \times 100)) = 100$	CRITICAL
bribery	Market Bribery	9	$\min(100, \text{int}(9 \times 1.15 / 10 \times 100)) = 100$	CRITICAL
secrecy	Secrecy / Info Leak	8	$\min(100, \text{int}(8 \times 1.15 / 10 \times 100)) = 92$	CRITICAL
employee_ethics	Employee Ethics	7	$\min(100, \text{int}(7 \times 1.15 / 10 \times 100)) = 80$	CRITICAL
change_in_comm	Change in Comms	6	$\min(100, \text{int}(6 \times 1.15 / 10 \times 100)) = 69$	HIGH
complaints	Complaints	4	$\min(100, \text{int}(4 \times 1.15 / 10 \times 100)) = 46$	MEDIUM

** Alone score assumes `high_confidence_bonus` ($\times 1.15$) applied since confidence is $1.0 \geq 0.80$ threshold*

5. How to Tune the Scoring

Because all parameters live in `compliance_matrix.yaml`, a compliance officer can change the scoring model without touching any Python code.

To change this...	Edit this YAML entry	Effect
Make complaints score higher	<code>complaints: base_weight: 7 # was 4</code>	Alone score rises from 46 to 80 (CRITICAL)
Make multi-category less aggressive	<code>multi_category_bonus: 1.05 # was 1.20</code>	Two-category emails get 5% boost instead of 20%
Lower the CRITICAL threshold	<code>priority_bands: critical: 70 # was 80</code>	Scores ≥ 70 now classified CRITICAL
Require higher confidence for HIGH alert	<code>confidence_thresholds: high: 0.90 # was 0.80</code>	Only very certain findings trigger HIGH alert
Add a new category	<code>categories: insider_trading: base_weight: 10</code>	New category available for the AI to detect

Important: The YAML categories section also feeds the AI Agent prompt. If you add a new category to the YAML, you must also add its ID and description to the `SYSTEM_PROMPT` in `compliance_agent.py` so the model knows to look for it.